

MERIT: Retrieval Is Not Contribution — Counterfactual Credit Assignment for Self-Evolving Agent Memory

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Abstract

Self-evolving language-model agents distill interaction trajectories into a persistent memory bank, and the quality of that self-improvement loop depends on how credit is assigned to individual memories. We argue that the credit signals used by current systems are *correlational*: a memory that merely co-occurs with success is reinforced, so *retrieval is mistaken for contribution*. We hypothesize that this reinforces *superstitious memories*—entries with zero or negative causal effect that nonetheless accumulate utility and occupy retrieval slots—and that their fraction grows with deployment length. The difficulty is that a memory’s causal value is defined only counterfactually, by how much outcomes degrade when it is removed, and per-memory intervention appears unaffordable in an online, non-stationary, closed loop. We propose MERIT, which is designed to make counterfactual credit affordable through a strictly linear pipeline: Randomized Interventional Trials (RIT) collect small-budget causal labels, Amortized Counterfactual Attribution (ACA) compresses them into an $O(1)$ -per-event score, and two minimal consumers—threshold governance and scope-gated retrieval—read the same signal. We plan a systematic evaluation testing whether ACA recovers causal credit at low cost, whether its consumers flatten superstition and reduce cross-task interference, and whether MERIT improves average success over the strongest baseline; the planned headline gain is [TBD], pending verification.

Introduction

Language-model agents are moving from stateless responders to experience-driven systems that improve over time by distilling their own interaction trajectories into a persistent memory bank and reusing it through a retrieve–execute–write loop [TBD]. The implicit premise of this paradigm is that the system knows *which memory deserves credit*: writing, merging, evicting, and retrieving all depend on a per-memory utility signal. Yet the signals in widely used systems are, on the characterization we adopt from prior descriptions and subject to bibliographic verification, *correlational*. When a task succeeds and a memory was retrieved, that memory’s utility is incremented. This conflates three distinct events:

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a memory can be *retrieved* without being *used*, and used without providing genuine *causal* help.

Retrieval is not contribution. A memory’s value is defined only counterfactually—by how much task outcomes degrade when it is removed from the retrieved set. Correlational credit does not measure this quantity. We hypothesize that it behaves like the reinforcement schedule in Skinner’s superstition experiment [TBD]: memories that merely co-occur with success are persistently strengthened, producing *superstitious memories*—entries with zero or negative causal contribution that nonetheless rise into the top-utility set and occupy retrieval slots. Because retrieval probability tends to track query solvability and high utility feeds back into higher retrieval rank, we hypothesize a self-reinforcing loop in which the fraction of superstitious memories *grows* with deployment length. A second, related failure is cross-task interference: a memory that is genuinely useful for one task family is indiscriminately retrieved into another and causes harm.

Why prior signals are insufficient. Existing approaches in our comparison commonly use observational, co-occurrence credit: success-trajectory distillation [TBD], environment-feedback utility [TBD], and Monte-Carlo or reinforcement updates over outcomes [TBD]. What is learned is the management *policy*; the underlying signal remains correlational. Counterfactual data-valuation methods such as Data Shapley [TBD], influence functions [TBD], and TracIn [TBD] target static, offline training data, while context-attribution methods [TBD] explain a single generation. None, to the best of our current and still-to-be-verified survey, deliver counterfactual credit in an online, non-stationary, closed loop where attribution itself changes the data distribution.

Approach. We propose MERIT, which replaces correlational bookkeeping with counterfactual credit while keeping it affordable. MERIT adds three tightly coupled components to the standard loop, in a strictly linear dependency $RIT \rightarrow ACA \rightarrow$ consumers: (i) *Randomized Interventional Trials* (RIT), a small-budget paired leave-one-out protocol that produces causal labels; (ii) *Amortized Counterfactual Attribution* (ACA), an $O(1)$ -per-event attributor trained on those labels and periodically recalibrated to track closed-loop drift; and (iii) two deliberately simple consumers of the resulting score—threshold-based credit governance and scope-gated

retrieval. Figure 1 contrasts correlational credit with this pipeline. Our central position is that *causal credit—not better heuristics—is the missing primitive of self-evolution.*

Contributions.

- **(C1) Superstitious memories.** We formalize and plan to quantify a diagnostic phenomenon: correlational credit breeds zero/negative-contribution memories in the top-utility set whose fraction rises with deployment time, and we introduce an interventional audit (RIT) with two statistics—the credit–contribution correlation (CCC) and the superstition rate (SR@ k)—to measure it.
- **(C2) Amortized counterfactual attribution.** We introduce ACA, which amortizes interventional labels into an $O(1)$ -per-event contribution score with periodic recalibration, and we plan to test whether it recovers causal credit substantially better than the correlational baseline at low cost.
- **(C3) Directly consumable causal credit.** We contribute a design in which a single causal-credit signal is directly consumable: threshold governance and scope-gated retrieval are two readouts of the same score. Governance, scope gating, and a per-operation reward substitution are evaluated as consumers of this one signal, not as separate methods, and we plan to test whether they respectively flatten superstition accumulation and reduce cross-task interference.
- **(C4) Systematic evaluation.** We plan a systematic evaluation across four benchmark categories, three backbones, and three seeds, testing whether MERIT improves average success over the strongest baseline at low token overhead.

Planned evidence. Every empirical statement in this paper is a planned test, not a measured result; unresolved quantities are written as registered ledger placeholders. A diagnostic pilot (Section 4) tests whether superstition accumulates; the mechanism analysis (Section 6) tests whether ACA recovers the audited signal in closed loop under the same axes as the pilot; and the main study tests whether the recovered signal translates into a planned average gain of [TBD] over the strongest baseline. Each claim carries an explicit falsification gate, and results that fail a gate will narrow the corresponding claim rather than be suppressed.

Related Work

We organize prior work by the credit signal it uses, not paper by paper, and position MERIT against each family. All bibliographic metadata is pending verification, and we avoid unqualified priority claims.

Self-evolving memory and experience distillation. A first family equips agents with a persistent memory that is written and reused across tasks: general memory stores [TBD], agentic memory with linked notes [TBD], workflow memory [TBD], reasoning-trajectory banks [TBD], skill or exemplar memory [TBD], and evolving stores [TBD]. On the characterization we adopt from their descriptions (pending verification), the credit these systems attach to a memory is

heuristic or co-occurrence based: a retrieved memory inherits the outcome of the task it accompanied. MERIT differs by auditing the causal validity of that credit with interventional evidence, rather than proposing another heuristic for accumulating it.

Learned memory management. A second family learns the memory-management policy with reinforcement or Monte-Carlo signals: reward-driven managers [TBD], policy-optimized memory agents [TBD], value-learned stores [TBD], and skill-management policies [TBD]. What is learned here is *which action to take*, while the reward that drives learning remains an outcome-level, correlational quantity. Complementary work improves write-time quality through multi-agent consensus [TBD], which is orthogonal to and composable with fixing full-lifecycle credit. MERIT targets the signal rather than the policy: it replaces the reward itself with a counterfactual score, and we later test whether substituting that score into such a manager lifts its performance.

Data valuation and context attribution. A third family measures the influence of data or context on model behavior: Data Shapley [TBD], influence functions [TBD], and TracIn [TBD] value static training points offline, while context attribution [TBD] explains a single generation. These methods share MERIT’s counterfactual framing but operate in a static, offline, or single-shot regime. The setting we address is online, non-stationary, and closed-loop: the attribution signal changes the retrieval distribution, which in turn changes future labels. This is what motivates amortization and periodic recalibration in ACA.

Interference and orthogonal directions. Cross-task interference in mixed-stream deployment has been observed [TBD] [TBD], and test-time reasoning methods [TBD] [TBD] improve behavior along an axis orthogonal to memory credit; we treat the latter as compatible baselines. The capability comparison in Table 5 summarizes where MERIT is intended to sit relative to these families; every competitor cell remains unresolved until its cited source is checked item by item, and an unresolved cell must not be read as a lack of capability.

Problem Formulation and Superstitious-Memory Hypothesis

Setting and notation. An agent processes a stream of tasks $\{(q_t, r_t)\}$, where q_t is a query or task and $r_t \in [0, 1]$ is an outcome signal (an environment success indicator or a consistency-checked judge score). It maintains a memory bank $\mathcal{M}_t = \{m_i\}$, a retrieval operator $R(q_t) \subseteq \mathcal{M}_t$ returning a top- k set, and a policy $\pi(y \mid q, R(q))$ that produces an output y . MERIT adds an attributor g_θ , a per-memory running credit φ_i , and a scope representation S_i for each memory. These eight objects are the only state the method reads or writes.

Counterfactual contribution. We define a memory’s value by intervention rather than by association.

Fig. 1 placeholder: Problem and solution schematic

Left panel: correlational credit and a top-utility memory set, with memories of non-positive causal contribution highlighted.
 Right panel: the $R_{IT} \rightarrow ACA \rightarrow$ consumers pipeline on the retrieve-execute-write loop.
 Future source: pilot $SR@k$ and mechanism analysis. No result geometry is drawn at this stage.

Figure 1: Planned hero schematic contrasting correlational credit with the counterfactual-credit pipeline. The eventual empirical annotation remains TBD until verification.

Method	Causal signal	O(1) online	Recalibration	Scope	Audit
Data Shapley	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Influence Functions	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
TracIn	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ContextCite	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ReasoningBank	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
SkeMex	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Memory-R1	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MemRL	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MERIT (ours)	yes	yes	yes	yes	yes

Table 5: Method-positioning skeleton. Every competitor cell remains TBD until its cited source is checked item by item; TBD must not be interpreted as lack of capability.

Definition 1 (Counterfactual contribution) *The contribution of memory m_i to task q is*

$$\varphi_i(q) \triangleq \mathbb{E}[r \mid q, R(q)] - \mathbb{E}[r \mid q, R(q) \setminus \{m_i\}], \quad (1)$$

where the expectation is over policy and environment randomness.

$\varphi_i(q)$ is the single-point interventional effect of removing m_i from the retrieved set. The full inter-memory value is a Shapley value over the retrieved set, of which single-point leave-one-out is the first-order approximation; the quality of that approximation is characterized in the boundary study (H5). The signal used by prior systems is, by contrast, associational.

Definition 2 (Co-occurrence utility) *The baseline signal is*

$$\hat{U}(m_i) = \frac{\sum_t \mathbf{1}[m_i \in R(q_t)] r_t}{\sum_t \mathbf{1}[m_i \in R(q_t)]}, \quad (2)$$

the average outcome of tasks in which m_i was retrieved.

Why correlational credit can be superstitious. We state the central analytical commitment as an informal proposition; a formal statement and proof are deferred to the appendix, and it is treated as a hypothesis until that proof is completed and checked.

Proposition 1 (Superstition bias, informal) *If retrieval events are correlated with the baseline solvability of the query (i.e., the covariance between retrieval and the memory-free success of q is nonzero), then $\hat{U}$ is an asymptotically biased estimator of φ . Moreover, under positive*

feedback in which utility participates in retrieval ranking, a stable superstition equilibrium can exist, in which $\hat{U}$ is high while $\varphi = 0$.

Semantic retrieval naturally induces the correlation condition, since easier queries are both more often solved and more often matched to stored memories. We make no claim beyond this statement; Proposition 1 remains a hypothesis pending its formal proof.

Assumptions. Three assumptions scope the method and are stated for rebuttal. To avoid collision with the ablation labels A1–A7 used later, we name the assumptions AS1–AS3.

- **AS1 (observable outcome).** Each task admits a usable r_t , either an environment success signal or a consistency-checked judge score.
- **AS2 (local replayability).** A task subset can be replayed from a fixed initial state (e.g., container snapshots or text environments). Only R_{IT} label collection requires AS2; ACA online scoring does not.
- **AS3 (low-order interaction).** Interaction effects within a retrieved set are bounded, so leave-one-out error against the Shapley value is controllable. Cases that violate AS3 (redundant memory pairs) are surfaced in the boundary study with a group-intervention fallback.

Objectives. ACA learns g_θ by minimizing a regression loss on R_{IT} labels, $\mathbb{E}[(g_\theta(x_i(q)) - \hat{\varphi}_i(q))^2]$, where $x_i(q)$ is a feature vector for the event. The system objective maximizes streaming cumulative success $\sum_t r_t$ subject to a token budget: retrieval plus R_{IT} sampling stay within $1 + \varepsilon$ times the

Fig. 2 placeholder: Baseline diagnostic

(a) x : system utility; y : counterfactual contribution ϕ .

(b) x : task index t (100–500); y : SR@20%.

Methods: ReasoningBank and MemRL.

The axis ranges and statistical definitions are locked to Fig. 5.

Figure 2: Planned diagnostic of credit-contribution correlation and superstition rate over deployment. All plotted values remain unresolved pilot outputs.

baseline cost, with $\varepsilon = [TBD]$ as a provisional hyperparameter.

Pilot and Diagnostic Study

This section diagnoses the failure that MERIT is designed to address; it introduces no second method. Every plotted quantity is an unresolved pilot output, and the study is governed by a prespecified falsification gate.

Systems and streams. We reproduce two credit designs on a unified codebase—a heuristic-credit bank in the style of ReasoningBank $[TBD]$ and a Monte-Carlo-learned bank in the style of MemRL $[TBD]$ —using the same backbone and retriever. We run them on two replayable streams (a text embodied environment and a multi-hop retrieval task), which are required because interventions must re-run paired rollouts from an identical initial state (AS2). Stream length and seed count are fixed as a planned configuration.

Interventional protocol. This is the first appearance of RRT. At a set of snapshot points along deployment ($t \in \{100, 200, 300, 400, 500\}$), we sample retrieval events and, for each retrieved memory, perform a paired leave-one-out comparison: the full retrieved set versus the same set with the memory removed, holding remaining context order fixed and inserting a filler placeholder so that position is not confounded. Each comparison uses $K = 5$ fixed-seed rollouts, and $\hat{\varphi}_i(q)$ is the resulting success-rate difference. Rollout counts and snapshot points are a planned configuration; the total rollout budget $[TBD]$ and token cost $[TBD]$ are unresolved and will be reported in an appendix cost table.

Statistics. We define two statistics parameterized by the ranking signal s , so that the same definitions apply to the baseline ($s = \hat{U}$) and to MERIT ($s = \hat{\varphi}$). The *credit-contribution correlation* $CCC(s, \varphi)$ is the Spearman correlation between the signal and the RRT contribution. The *superstition rate* $SR@k(s)$ is the fraction of the top- $k\%$ memories by s whose contribution is non-positive ($\varphi \leq 0$). Figure 2 plots the baseline CCC and the SR@20% trajectory over deployment.

Planned readings and controls. We test whether the baseline correlation is low ($[TBD]$ for the heuristic bank and

$[TBD]$ for the learned bank) and whether SR@20% rises from $[TBD]$ at $t = 100$ to $[TBD]$ at $t = 500$, as predicted if correlational credit does not self-correct. Two controls guard against alternative explanations. A split-half re-test of $\hat{\varphi}$ gives reliability $[TBD]$, testing whether a low correlation is a measurement artifact. Removing the entire $\varphi > 0$ subset and measuring the success drop $[TBD]$ tests whether memory has value at all, so that a defective credit assignment—rather than useless memory—is the operative problem.

Falsification gate. If the credit-contribution correlation exceeds 0.5, or if the superstition rate does not rise with deployment, the attribution hypothesis is not supported; we will then narrow the claim to heterogeneous streams rather than force a positive reading. The diagnostic yields two requirements for the method: approximate the RRT credit online, and expose the same credit signal to downstream consumers. Both are realized by MERIT, whose architecture is shown in Figure 3.

Method: MERIT

ACA predicts the same contribution φ that the pilot audits in Figure 2; it is an amortization of that target, not a new quantity. MERIT adds three components to the standard retrieve-execute-write loop in a strictly linear dependency, $RRT \rightarrow ACA \rightarrow$ consumers, with no parallel branches and no unrelated modules (Figure 3). All thresholds and coefficients below are provisional hyperparameters.

Randomized Interventional Trials

RRT continuously produces causal labels at low budget. With sampling probability $p = [TBD]$ (a provisional hyperparameter; the source default is 5%), an incoming replayable task enters the trial group. For each memory in its retrieved set, RRT performs the paired leave-one-out comparison of Section 4 with $K = 5$ fixed-seed rollouts and the same position-control filler, and emits a quadruple $(q, m_i, \text{trajectory}, \hat{\varphi}_i)$ into a label pool. Sampling runs only on tasks that satisfy AS2. Because the naive alternative—letting the agent self-judge whether a memory helped—can reproduce a self-confirmation trap, we retain it only as the ablation variant A-judge rather than as the label source.

Amortized Counterfactual Attribution

ACA compresses the interventional labels into an $O(1)$ -per-event score. The feature vector $x_i(q)$ has three groups: a query representation, a memory representation (a frozen embedding encoder), and *usage-behavior* features that distinguish a memory that was retrieved but ignored from one that was actually used— n -gram and edit overlap between memory content and output, explicit citation markers, context position and retrieval rank, and the log-likelihood gain on memory tokens. A two-layer perceptron regresses $\hat{\varphi} \in [-1, 1]$ under a Huber loss ($\delta = 0.1$, provisional). Because the consumers change the closed-loop retrieval distribution, ACA performs *periodic recalibration* every 100 tasks (planned configuration), fine-tuning on the newest labels and applying isotonic calibration. At inference, ACA scores each of the

Fig. 3 placeholder: MERIT architecture

Retrieve-execute-write loop with three ordered additions: paired-LOO Rrr labels, amortized ACA scores with recalibration, and two consumers (credit governance and scope-gated retrieval). This is a data-free schematic and can be drawn before experiments finish.

Figure 3: Planned architecture of the strictly ordered Rrr $\rightarrow$ ACA $\rightarrow$ consumers pipeline.

top- k retrieved memories and updates an exponential moving average of the running credit $\bar{\varphi}_i$ and the positive/negative query prototype sets. The trainable parameter count is [TBD].

Two consumers of one signal

Governance and scope gating are two readouts of the same $\bar{\varphi}$; neither is an independent method.

Credit governance. Three threshold rules update retention. A memory is *evicted* when $\bar{\varphi}_i < -0.02$ and its sample count $n_i \geq 8$; two memories are *merged* when their embedding similarity exceeds 0.9 and both have $\bar{\varphi} > 0$, combining their credit by weighting; and new memories are *quarantined*, receiving a UCB-style exploration bonus over their first $n_{\min} = 3$ retrievals to avoid cold-start death without contributing a ranking bonus. The ± 0.02 dead-zone is deliberate, because the sign of the estimate is least stable near $\varphi \approx 0$. Threshold rules—rather than a learned policy—are used so that the ablation A2, which swaps the causal score back to co-occurrence utility under the same rules, isolates the contribution of the *signal*.

Scope-gated retrieval. Each memory maintains positive and negative query prototypes (the mean embedding of queries with $\hat{\varphi} > \tau^+$ and $\hat{\varphi} < \tau^-$). The retrieval score combines semantic relevance, scope agreement, and credit,

$$\text{score} = \alpha \cdot \text{rel} + \beta (\text{sim}(q, \text{proto}^+) - \text{sim}(q, \text{proto}^-)) + \gamma \cdot \bar{\varphi}_i, \quad (3)$$

with $(\alpha, \beta, \gamma) = (1.0, 0.5, 0.3)$ as provisional defaults. This is intended to address cross-task interference by learning *where* each memory applies. The same $\hat{\varphi}$ can also replace a manager’s outcome reward, which we evaluate later as a signal-portability check rather than a separate contribution.

Cost and complexity

The dependency Rrr $\rightarrow$ ACA $\rightarrow$ consumers is a verified linear chain. ACA inference is $O(1)$ in bank size at a fixed feature dimension; Rrr label collection and periodic recalibration are amortized over the stream and accounted for separately. This is a design property: constant-time online scoring follows from the architecture, whereas the end-to-end token and memory overhead—[TBD] tokens and [TBD] of resident VRAM—and the per-event latency [TBD] remain planned measurements. Algorithm 1 summarizes the loop.

Algorithm 1 MERIT main loop

Input: stream $\{q_t\}$, bank $\mathcal{M}$, retriever R , policy π , attributor g_θ , Rrr budget p

- 1: **for** $t = 1, 2, \dots$ **do**
- 2: $C \leftarrow R(q_t)$ {scope-gated: $\alpha \cdot \text{rel} + \beta \cdot \text{scope} + \gamma \cdot \text{credit}$ }
- 3: $y_t, \text{traj} \leftarrow \pi(q_t, C)$; $r_t \leftarrow \text{env/judge}(y_t)$
- 4: **for** $m_i \in C$ **do**
- 5: $\bar{\varphi}_i \leftarrow g_\theta(x_i(q_t, \text{traj}))$; update $\bar{\varphi}_i$ and prototypes
- 6: **end for**
- 7: **if** $\text{replayable}(q_t)$ **and** $\text{rand}() < p$ **then**
- 8: **for** $m_i \in C$ **do**
- 9: label $\leftarrow$ paired-LOO($q_t, C, m_i, K=5$); add to pool
- 10: **end for**
- 11: **end if**
- 12: governance($\mathcal{M}$): evict / merge / quarantine by rules on $(\bar{\varphi}, n)$
- 13: **if** $t \bmod 100 = 0$ **then**
- 14: recalibrate g_θ on pool (Huber + isotonic)
- 15: **end if**
- 16: **end for**

Experiments and Analysis

This section specifies protocols, table skeletons, figure placeholders, and conditional interpretation plans. No ranking or numerical conclusion is asserted before verification; every result cell is an unresolved ledger placeholder, and each analysis states the test it will run and the reading that would support or fail its hypothesis. Table 4 maps each hypothesis to its metric, experiment, and claim.

Setup

We plan to evaluate three backbones—a primary open model, a larger open model as a scale-trend point, and a closed model as a generalization row—on a unified memory-system code-base for baseline fairness. ACA uses a frozen embedding encoder with a two-layer head. Evaluation spans four benchmark categories: streaming task streams (dialogue and agentic), a web-agent benchmark, a long-term memory benchmark, and a conversational memory benchmark; the pilot uses two replayable environments. For fairness, all methods share the backbone, retriever, and top- $k = 4$ budget, with baselines given an equal token budget for their own mech-

Hyp.	Metric	Experiment	Claim	Status
H1	SR@20% slope	A4 / mechanism	C3	HYPOTHESIS
H2	Held-out CCC	Fig. 5a / calibration	C2	HYPOTHESIS
H3	CTI; transfer CCC	A3 / transfer	C3	HYPOTHESIS
H4	AVG versus token cost	Efficiency study	C4	HYPOTHESIS
H5	Bank / heterogeneity / redundancy	Boundary study	C4	HYPOTHESIS

Table 4: Hypothesis-to-experiment map. This table contains design metadata rather than measured results.

anism; we report mean and standard deviation over seeds $\{13, 42, 2026\}$ with paired-bootstrap significance and per-point confidence bands. A subset uses a cross-source judge, of which 200 examples are human-checked for agreement $[TBD]$. A single main run costs $[TBD]$, with the full cost table in the appendix.

Main results

Table 1 orders methods by lineage—no-memory, heuristic credit, learned credit, orthogonal reasoning, and an intervention reference—and orders the stream columns S1–S5 by increasing heterogeneity; the identities of S2–S4 are not specified in the source. We test three predictions against the eventual numbers: whether the gain grows with stream heterogeneity (from $[TBD]$ on the single-domain stream to $[TBD]$ on the mixed stream); whether learned-credit baselines are reliably better than heuristic ones; and whether MERIT reaches $[TBD]$ of the RIT-Full reference gain at a fraction of its cost. The headline average gain over the strongest baseline is $[TBD]$. The RIT-Full row is an intervention reference and is treated as a ceiling only for quantities where that status follows by construction.

Signal-portability check: reward substitution

To test whether the improvement originates at the signal layer rather than in a particular pipeline, we substitute $\hat{\varphi}$ for a manager’s original outcome reward, leaving its optimization otherwise unchanged (Table 2). We test whether this substitution lifts two reinforcement-style managers by $[TBD]$ and $[TBD]$ respectively. This is a portability check on one signal, not a separate method; the full table is a candidate for the appendix.

Ablation

Table 3 removes one component at a time and adds two alternative-signal controls. We read it as a causal chain—credit fidelity (CCC) $\rightarrow$ mechanism and interference change (SR@20%, CTI) $\rightarrow$ task outcome (AVG)—rather than as isolated cells. The pivotal test is A2: swapping the causal score back to co-occurrence utility under identical governance rules. If the gain persists under A2, the contribution is not localized to the signal, and the interpretation must be revised.

Mechanism and calibration

Figure 5 reuses the axes and statistical definitions of the baseline diagnostic (Figure 2) so that the two are directly

Fig. 5 placeholder: Mechanism recovery

- (a) x : predicted contribution $\hat{\phi}$; y : RIT contribution ϕ .
(b) x : task index t (100–500); y : SR@20%.

Compare ACA (MERIT) with the correlational baseline. Axis ranges and statistical definitions must be identical to Fig. 2.

Figure 5: Planned mechanism-closure analysis comparing attribution fidelity and superstition-rate trajectories under the same definitions and axes as the baseline diagnostic.

comparable. Panel (a) tests whether ACA’s held-out credit-contribution correlation $[TBD]$ improves on the baseline $[TBD]$; the source gate for H2 is a Spearman correlation of at least 0.6. Panel (b) tests whether the superstition rate is flattened to $[TBD]$, a reduction of $[TBD]$ relative to the baseline at $t = 500$; the H1 reading is the *slope* of this trajectory, compared between the full method and the no-governance ablation across the deployment checkpoints. Calibration is assessed with a held-out reliability diagram and expected calibration error $[TBD]$, per-domain calibration, and a split-half re-test, with particular attention to sign instability near $\varphi \approx 0$.

Boundary and scaling

We test whether the gain holds under a larger backbone (from $[TBD]$ to $[TBD]$, with stronger baselines but an unchanged superstition mechanism) and whether an attributor trained on the pilot environments transfers zero-shot, measured by the transfer correlation $[TBD]$ (H3). For the boundary (H5), we plan three trend lines: gain versus bank size (the small-bank bin has planned gain $[TBD]$), gain versus stream heterogeneity (single-domain gain $[TBD]$), and gain versus injected redundancy. The redundancy trend has no registered result identifier and is reported as an EVIDENCE GAP, kept exploratory; the hypotheses are that leave-one-out underestimation grows with redundancy and that the group-intervention variant (A7) mitigates it.

Efficiency

Figure 4 plots cumulative success over the stream and, as an inset shown only after verification, a planned RIT-budget sweep over $p \in \{1, 2, 5, 10\}\%$; the inset is labeled a

Method	S1	S2	S3	S4	S5	W-A	LME	LCM	AVG
No-Memory ReAct	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Full-History Stuffing	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Mem0	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A-MEM	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AWM	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ReasoningBank	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
SkeMex	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Memory-R1	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Mem- α	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MemRL	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Reflexion	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ReasoningBank+MaTTS	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MERIT (ours)	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
RIT-Full reference	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 1: Main-results skeleton with 14 method rows. S1–S5 are ordered by planned stream heterogeneity; the identities of S2–S4 remain unspecified in the source. All result cells are unresolved.

Manager / reward	AVG	Δ original
Memory-R1 / original	<i>TBD</i>	–
Memory-R1 / $\hat{\phi}$	<i>TBD</i>	<i>TBD</i>
Mem- α / original	<i>TBD</i>	–
Mem- α / $\hat{\phi}$	<i>TBD</i>	<i>TBD</i>

Table 2: Reward-substitution skeleton. The table will test signal portability; it does not yet establish that any improvement originates at the signal layer.

Fig. 4 placeholder: Streaming and efficiency

Main panel: x = task index; y = cumulative success. Compare MERIT, strongest verified baselines, and the RIT-Full intervention reference.
Optional inset: planned RIT-budget p sweep, shown only after data verification.

Figure 4: Planned test of whether performance differences concentrate in the later portion of the stream and whether the RIT budget reaches a Pareto-efficient region.

planned comparison rather than a measured curve. We test whether performance differences concentrate in the later, most-polluted portion of the stream—what share *[TBD]* of the average gain arises there—and whether the RIT budget saturates near the source default, placing MERIT at a Pareto-efficient knee (H4). Table 6 records relative token cost, success, latency, and memory overhead against the co-occurrence baseline and the intervention reference.

Case study

Figure 6 plans two qualitative timelines. In the success case, a useful multi-hop memory and a superstitious “retry the

Fig. 6 placeholder: Case study and AS3 boundary

Timeline A: useful memory versus superstitious memory, with utility, RIT credit, scope, and governance events.
Timeline B: complementary-memory failure and the group-intervention fallback.
Future source: verified case-study event logs.

Figure 6: Planned qualitative analysis of a successful governance case and a complementary-memory failure that exposes the AS3 boundary.

same query” memory are tracked over deployment; we test whether the latter’s inflated utility is assigned near-zero RIT credit and leads to eviction at task *[TBD]*, while the former’s scope is characterized correctly. In the failure case, two complementary memories each hold half of a procedure; we test whether individual leave-one-out is near zero for both so that governance nearly removes both, and whether the group-intervention fallback recovers their joint credit—an honest display of the AS3 boundary. We expect a share *[TBD]* of MERIT’s remaining failures to stem from the bank simply lacking relevant experience, which no credit signal can repair.

Limitations and Conclusion

Limitations. MERIT inherits three scope boundaries. It depends on local replayability (AS2): fully non-replayable, single-shot environments cannot yield RIT labels, and world-model-simulated replay or off-policy evaluation would be needed as a substitute. It depends on bounded interaction (AS3): leave-one-out underestimates credit under high redundancy, and the group-intervention fallback adds cost. And its benefit has a boundary (H5): single-domain short streams and small banks are expected to yield little gain,

Variant	AVG	Δ AVG	SR@20% ($t = 500$)	CCC	CTI
Full MERIT	<i>TBD</i>	0	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A1: no recalibration	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A2: co-occurrence signal	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A3: no scope gating	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A4: no governance	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A5: no usage features	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A-judge: LLM self-judge	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A7: group / sampled Shapley	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 3: Ablation skeleton. The table will test credit fidelity, mechanism statistics, interference, and task outcomes without presupposing that any component is necessary.

Configuration	Relative tokens	AVG	ACA ms/event	Extra VRAM	RIT overhead
Co-occurrence baseline	$1 \times$	<i>TBD</i>	–	–	0
MERIT	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A1 (static attributor)	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
RIT-Full reference	<i>TBD</i>	<i>TBD</i>	–	–	<i>TBD</i>

Table 6: Efficiency skeleton. All non-reference values remain unresolved; no Pareto ranking is assigned before verification.

so MERIT is intended for long-term deployment rather than short sessions. We further scope the contribution to credit assignment for existing memories and leave write-time quality out of scope; MERIT composes with methods that address it. A rebuttal-sensitive risk is that, if attribution scores can be influenced by externally supplied content, the audit becomes a new attack surface for memory-poisoning “credit laundering,” which we flag for future defense. Whether ACA transfers across fundamentally different modalities beyond the tested text and agentic environments is not specified by our source and is left as an open question, excluded from C1–C4.

Conclusion. We argued that self-evolving agent memory is limited less by better heuristics than by a missing primitive: causal credit. Correlational bookkeeping mistakes retrieval for contribution and, we hypothesize, breeds superstitious memories that accumulate over deployment. MERIT is designed to supply counterfactual credit affordably by collecting small-budget interventional labels (RIT), amortizing them into an $O(1)$ -per-event score (ACA), and exposing that single score to two minimal consumers. The evaluation is planned rather than completed: it tests whether ACA recovers the audited signal in closed loop, whether its consumers flatten superstition and reduce cross-task interference, and whether the recovered signal improves average success at low overhead—each behind an explicit falsification gate. Two concrete directions follow: structure-aware interaction attribution to relax AS3, and replay substitutes to relax AS2.